Git & GitHub Team Collaboration Handbook

Step-by-Step Workflow Guide: Branching, Commits, PRs & Zero Conflicts (Hinglish)

GOLDEN RULE (0% Conflicts): Repository me 3 main folders hain: **Frontend/**, **backend/**, aur **Agent/**. Agar Frontend dev sirf Frontend/ me code kare, Backend dev sirf backend/ me, aur Agent dev Agent/ me - toh **99% kabhi koi merge conflict aayega hi nahi!** Koshish karein ki dusre ke folder ki files me bina discuss kiye changes na karein.

1. First-Time Setup on Teammate Laptop (One-Time Only)

Project shuru karte waqt har teammate ko bas ye 2 kaam karne hain:

- **Step A:** GitHub repo invite accept karein (Lead aapko GitHub repo ka collaboration invite bhejega).
- **Step B:** Apne computer par Terminal / PowerShell khol kar repo clone karein:

```
git clone https://github.com/HarshitD0501/<REPO_NAME>.git
cd <REPO_NAME>
```

2. Branch Banana (Apna Personal Workspace)

Kabhi bhi seedha 'main' branch me code mat karo! Har teammate apni dedicated feature branch banayega:

- Nayi branch banane aur turant usme shift hone ka command: git checkout -b <branch-name>

```
# Frontend developer:
git checkout -b frontend-dev

# Backend developer:
git checkout -b backend-dev

# AI / Voice Agent developer:
git checkout -b agent-dev
```

Branch check kaise karein? Run karein git branch. Jis branch ke aage *** (star)** laga ho, aap abhi usi me khade hain.

3. Daily Coding & Commit Control (Commit Kahan Jayega?)

Sawal: Agar main commit karta hoon, toh wo main me jayega ya meri banayi hui branch me?
Jawab: Git hamesha usi branch me commit karta hai jisme aap **currently khade hain**. Agar aap frontend-dev par hain, toh commit sirf frontend-dev me save hoga, main bilkul safe rahegi!

Apne changes save (commit) karne ke 3 simple steps:

```
# 1. Check karein kya-kya modify hua hai
git status

# 2. Saari files ko stage karein
git add .

# 3. Meaningful message ke saath commit karein
git commit -m "Added patient search component in Frontend"
```

4. GitHub par Push karna & VS Code 'Sync' Button

Jab aapne commit kar liya aur ab code GitHub par daalna hai:

- **Pehli baar apni branch push karte waqt:**

```
git push -u origin frontend-dev # (apni branch ka naam likhein)
```

- **Subsequent pushes:** Ek baar -u chalane ke baad agli baar aap seedha git push chala sakte hain.
- **VS Code Sync Button:** VS Code ke bottom bar me neela **'Sync Changes'** button hota hai. Us par click karne se wo automatically pull + push dono kar deta hai!

5. Pull Request (PR) & Merging into 'main'

- **PR kaun banayega?** Jis teammate ne code likha hai aur apni branch push ki hai.
- **GitHub par kya dikhega?** Push karte hi GitHub repo page par yellow notification bar aayega: *'frontend-dev had recent pushes'* aur saath me green button: **Compare & pull request**.
- **PR kaise raise karein?** Green button click karein -> Chhota title likhein (e.g. 'Added API endpoints') -> **Create pull request** par click karein.
- **Merge kaun karega?** Team Lead ya koi bhi teammate PR check karega aur neeche green button dabayega: **Merge pull request** -> **Confirm merge**. Code ab main me jud gaya!

6. Dusron ka Code Apne Paas Lana (Sync with Latest main)

Kyu zaroori hai? Jab Backend wale ne naye APIs banaye aur unka PR main me merge hua, toh wo code aapki working branch me laana zaroori hai taaki aap un APIs ko use kar sakein aur future merge conflict se 100% bach sakein.

- **Option A: 1-Line Pro Shortcut (Sabse Fast & Recommended):**

Apni branch (jaise frontend-dev) me khade rehkar seedha ye ek command chalayein:

```
git pull origin main
```

(Yeh command GitHub ke main se latest code download karke seedha aapki current working branch me merge kar deti hai!)

- **Option B: Manual 4-Step Tarika (Concept samajhne ke liye):**

```
# 1. Main me jayein aur GitHub se download karein:
git checkout main
git pull origin main

# 2. Wapas apni branch me jayein aur main ka code apni branch me daalein:
git checkout frontend-dev
git merge main # <- Isse main ka naya code aapki branch me jud jata hai!
```

7. Merge Conflicts: Kyu Aate Hain aur Aaram Se Kaise Solve Karein?

Conflict kab aata hai? Jab do log **ek hi file ki bilkul same line** ko alag-alag tarike se badal dete hain. Isse ghabrane ki bilkul zaroorat nahi hai!

- **Tareeqa 1 (GitHub Web Editor - Sabse aasan):** PR page par GitHub khud *'Resolve conflicts'* ka button deta hai. Us par click karein, jo sahi code rakhna ho wo rakhein, baaki <<<<<<<, =======, >>>>>>> wali lines delete karke **'Mark as resolved'** daba dein.
- **Tareeqa 2 (VS Code me):** Conflict aate hi VS Code file me 3 buttons dikhata hai: *[Accept Current Change]* | *[Accept Incoming Change]* | *[Accept Both]*. Bas ek click karein, file save karein, aur git commit karke push kar dein.

8. Quick Command Cheat-Sheet (Summary)

Task	Command	Description
Repo Clone	`git clone <repo-url>`	First time GitHub repo apne laptop par lana
Nayi Branch	`git checkout -b <name>`	Nayi branch bana kar usme switch hona
Active Branch	`git branch`	Dekhna ki aap abhi kis branch me hain (* star)
Branch Switch	`git checkout <branch>`	Dusri existing branch me jaana
Files Stage	`git add .`	Saare local changes ko commit ke liye stage karna
Save Commit	`git commit -m "msg"`	Changes ko local branch me save karna
First Push	`git push -u origin <branch>`	Nayi branch ko pehli baar GitHub par bhejna
Normal Push	`git push`	Agli baar ke commits GitHub par bhejna (or Sync button)
Quick Sync	`git pull origin main`	Apni branch me khade-khade latest main ka code merge karna